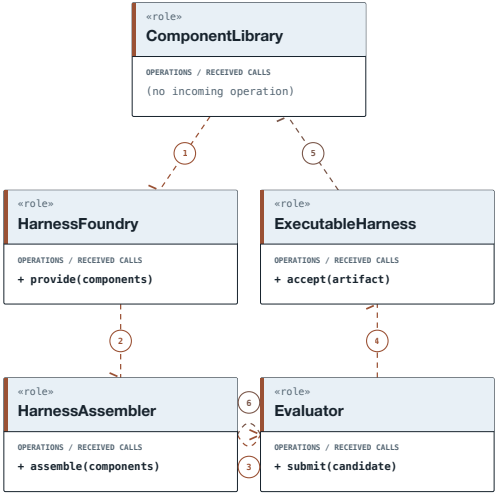
E05 Composable harness foundry

Evolution / UML CLASS VIEW / ANALYTICAL MODEL

Construct and evolve reusable harness components.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Harness behavior is assembled from independently typed components.
TRIGGER	Components are selected, composed, or replaced.
STRUCTURE	Component interface, composition builder, evaluator, and artifact registry.
EFFECT	The built harness is executable and its component provenance is recoverable.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Components have compatible contracts and composition operators.
ENSURES	Invalid composition is rejected before execution.
INVARIANT	Reusing a component does not silently reuse privileged state.
VIOLATION	Build conflicts, incompatible versions, and evaluation failure are explicit.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 ComponentLibrary → HarnessFoundry provide(components)
2 HarnessFoundry → HarnessAssembler assemble(components)
3 HarnessAssembler → Evaluator submit(candidate)
4 Evaluator → ExecutableHarness accept(artifact)
5 ExecutableHarness ↩ ComponentLibrary report(reusable)
6 Evaluator ↩ HarnessAssembler feedback(failure)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Typed processors, foundry DSL, component search, and immutable builders.
TEST_OBLIGATION	Test composition, incompatible component, rebuild, and provenance.
NEGATIVE_CONTROL	Separate helper functions without component contracts are insufficient.